

PRACTICAL 2

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create database week2
use week2
CREATE TABLE Books (
    book_id INT PRIMARY KEY,
    title VARCHAR(100) NOT NULL,
    isbn VARCHAR(20) UNIQUE,
    published_year INT CHECK (published_year < 2027)
);
INSERT INTO Books (book_id, title, isbn, published_year) VALUES
(1, 'The Great Gatsby', '9780743273565', 1925),
(2, 'To Kill a Mockingbird', '9780061120084', 1960),
(3, '1984', '9780451524935', 1949);
select * from books
CREATE TABLE Members (
    member_id INT PRIMARY KEY,
    full_name VARCHAR(100),
    email VARCHAR(100) UNIQUE
);

INSERT INTO Members (member_id, full_name, email) VALUES
(101, 'John Smith', 'john.smith@email.com'),
(102, 'Emma Wilson', 'emma.wilson@email.com'),
(103, 'Michael Brown', 'michael.brown@email.com');
SELECT * FROM Members;
CREATE TABLE Loans (
    loan_id INT PRIMARY KEY,
    member_id INT,
    book_id INT,
    loan_date DATE,
    FOREIGN KEY (member_id) REFERENCES Members(member_id),
    FOREIGN KEY (book_id) REFERENCES Books(book_id)
);
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
SELECT * FROM Loans;
SELECT m.full_name AS Member_Name, b.title AS Book_Title

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DBSE

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FROM Loans l
INNER JOIN Members m ON l.member_id = m.member_id
INNER JOIN Books b ON l.book_id = b.book_id;
SELECT published_year, COUNT(book_id) AS Total_Books
FROM Books
GROUP BY published_year
ORDER BY published_year;
CREATE TABLE Donation_History (
    donation_id INT PRIMARY KEY,
    book_id INT,
    donor_name VARCHAR(100),
    donation_date DATE,
    FOREIGN KEY (book_id) REFERENCES Books(book_id)
);
START TRANSACTION;
INSERT INTO Books (book_id, title, isbn, published_year)
VALUES (4, 'Animal Farm', '9780451526342', 1945);
INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
VALUES (1, 4, 'Raj Kumar', CURDATE());
COMMIT;
CREATE INDEX idx_books_isbn ON Books(isbn);
SELECT * FROM Books WHERE isbn = '9780451524935';
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book_id	title	isbn	published_year
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL